Advanced Statistics - Assignment 1

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You are given a dataset of integer-valued observations

$$D = \{x_1, x_2, \ldots, x_N\}$$

representing the number of events observed in identical time intervals. Analyze the data using the Bayesian framework following the subsequent points.

1

Assume that the data are generated from a Poisson distribution:

$$x_i \sim Poisson(\lambda)$$

with a Gamma prior on λ:

$$\lambda \sim Gamma(\alpha, \beta)$$

1.1

Derive the posterior distribution $p(\lambda|D)$.
We apply Bayes' theorem to get the posterior distribution:

$$p(\lambda|D) = \frac{p(D|\lambda)p(\lambda)}{p(D)} \tag{1}$$

The likelihood is given by the product of Poisson distributions, as the data is assumed to be independent and identically distributed (i.i.d.):

$$p(D|\lambda) = \prod_{i=1}^{N} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum_{i=1}^{N} x_i} e^{-N\lambda}}{\prod_{i=1}^{N} x_i!} \tag{2}$$

As $\sum_{i=1}^{N} x_i$ is a sufficient statistic for λ, we denote it as $y = \sum_{i=1}^{N} x_i$. The prior is a Gamma distribution:

$$p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \tag{3}$$

where $\alpha > 0$ and $\beta > 0$ are hyperparameters. Since we have no strong prior knowledge and are using the Gamma distribution primarily for its conjugate properties with the Poisson likelihood, we select an uninformative prior. By setting $\alpha = 0.001$ and $\beta = 0.001$, the prior approximates a flat distribution, ensuring the posterior is driven predominantly by the empirical data.

The evidence is given by the integral of the joint distribution over all possible values of λ:

$$\begin{aligned} p(D) &= \int_0^\infty p(D|\lambda)p(\lambda)d\lambda \\ &= \int_0^\infty \left(\frac{\lambda^y e^{-N\lambda}}{\prod_{i=1}^N x_i!}\right)\left(\frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}\right)d\lambda \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)\prod_{i=1}^N x_i!}\int_0^\infty \lambda^{(\alpha+y)-1}e^{-(\beta+N)\lambda}d\lambda \end{aligned} \tag{4}$$

where we can evaluate the last integral by noticing it is an unnormalized Gamma distribution, yielding the analytical evidence:

$$p(D) = \frac{\beta^\alpha\Gamma(\alpha+y)}{\Gamma(\alpha)(\beta+N)^{\alpha+y}\prod_{i=1}^N x_i!} \tag{5}$$

Finally, dividing the joint distribution by the evidence leads to the posterior distribution:

$$p(\lambda|D) = \frac{(\beta+N)^{\alpha+y}}{\Gamma(\alpha+y)}\lambda^{\alpha+y-1}e^{-(\beta+N)\lambda} \tag{6}$$

which is perfectly recognizable as a new Gamma distribution with updated hyperparameters $\alpha' = \alpha + y$ and $\beta' = \beta + N$. Therefore, it is known that the posterior mean and variance are respectively:

$$E[\lambda|D] = \frac{\alpha'}{\beta'} = \frac{\alpha+y}{\beta+N} \qquad \text{and} \qquad Var[\lambda|D] = \frac{\alpha'}{\beta'^2} = \frac{\alpha+y}{(\beta+N)^2} \tag{7}$$

As mentioned before, choosing prior hyperparameters close to zero results in a posterior mean and variance that are determined almost entirely by the observed data.

```r
library(ggplot2)

# Load the data into data frame
file <- "/home/dghezzi/AdvancedStatistics/ex1/dataset.txt"
data <- read.table(file, header = FALSE)
colnames(data) <- c("data")

# 1.a Derivation of posterior distribution
# Since as prior we have a Gamma distribution, the posterior distribution will also be a Gamma distribution.
# The parameters of the posterior distribution can be calculated as follows:

alpha_prior <- 0.001
beta_prior <- 0.001

alpha_post <- alpha_prior + sum(data$data)
beta_post <- beta_prior + nrow(data)
post_mean <- alpha_post / beta_post # mean of the posterior distribution
post_var <- alpha_post / (beta_post^2) # variance of the posterior distribution


cat(sprintf("The posterior distribution is Gamma with alpha = %.3f and beta = %.3f\n", alpha_post, beta_post))
cat(sprintf("The mean of the posterior distribution is: %.3f\n", post_mean))
cat(sprintf("The variance of the posterior distribution is: %.3f\n", post_var))
```

```
The posterior distribution is Gamma with alpha = 2670.001 and beta = 500.001
The mean of the posterior distribution is: 5.340
The variance of the posterior distribution is: 0.011
```

1.2

Compute the maximum a posteriori (MAP) estimate of λ.
The maximum a posteriori estimate selects the single most probable value in the posterior distribution:

$$\hat{\lambda}_{\text{MAP}} = \underset{\lambda}{\operatorname{argmax}}\, p(\lambda|D) \tag{8}$$

To solve this problem analytically, we set the first derivative of the unnormalized posterior with respect to λ equal to zero:

$$\frac{d}{d\lambda}\left[\lambda^{\alpha'-1}e^{-\beta'\lambda}\right] = 0 \tag{9}$$

Applying the product rule leads to the following equation:

$$e^{-\beta'\lambda}\lambda^{\alpha'-2}\left[(\alpha'-1)-\beta'\lambda\right] = 0 \tag{10}$$

Since $e^{-\beta'\lambda}$ is strictly positive and we are evaluating for $\lambda > 0$, the roots of this equation depend entirely on the bracketed term. As in our case we have $\alpha' > 1$, the solution of the equation is:

$$\hat{\lambda}_{\text{MAP}} = \frac{\alpha'-1}{\beta'} = \frac{\alpha+y-1}{\beta+N} \tag{11}$$

We can observe that the MAP estimator coincides to the mean of the posterior as $\alpha' \to +\infty$. Indeed, in the code we can see that these two values are almost the same.

```r
# 1.b Maximum a posteriori (MAP) estimate of lambda
# From analytical calculations, the MAP estimate of lambda can be calculated as follows:

map_estimate <- (alpha_post - 1) / beta_post
cat(sprintf("The MAP estimate of lambda is: %.3f\n", map_estimate))

# Plot of the posterior distribution and the MAP estimate
x <- seq(0, 10, length.out = 1000)
posterior <- dgamma(x, shape = alpha_post, rate = beta_post)
plot_data <- data.frame(lambda = x, density = posterior)
ggplot(plot_data, aes(x = lambda, y = density)) +
  geom_area(fill = "steelblue", alpha = 0.2) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_vline(xintercept = map_estimate, color = "red", linetype = "dashed", linewidth = 0.5) +
  labs(title = "Posterior Distribution",
       subtitle = paste("MAP Estimate of lambda =", round(map_estimate, 3)),
       x = expression(lambda),
       y = "Density") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold"))
ggsave("ex1/posterior_MAPestimate_Poisson.png", width = 8, height = 6, dpi = 300)
```

```
The MAP estimate of lambda is: 5.338
```

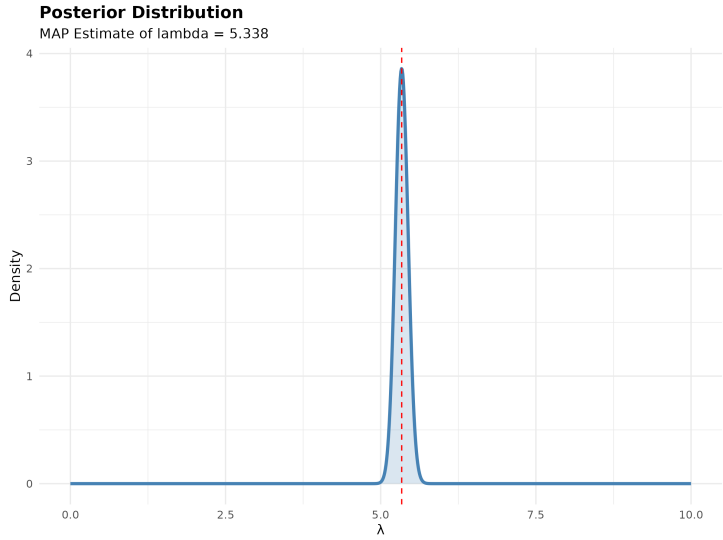

Figure 1: Posterior distribution for Poissonian inference and MAP estimator of λ. The posterior is unimodal and sharply peaked around the MAP estimator.

1.3

Discuss whether the posterior is approximately symmetric for the given dataset, and provide an adequate estimate of the uncertainty on λ.

To check the posterior symmetry, we rely on the skewness and excess kurtosis, which are analytically known for Gamma distributions:

$$\text{Skewness} = \frac{2}{\sqrt{\alpha'}} \qquad \text{Excess Kurtosis} = \frac{6}{\alpha'} \tag{12}$$

Because our updated shape parameter α' is large, both of these values are approximately zero. This confirms that the posterior is highly symmetric and sharply peaked. Hence, an adequate estimate of the uncertainty on λ is given by a Gaussian approximation. To provide it, we proceed as follows. First, we define the log-posterior $\mathcal{L}(\lambda)$:

$$\mathcal{L}(\lambda) = \log[p(\lambda|D)] = \alpha' \log \beta' - \log \Gamma(\alpha') + (\alpha' - 1) \log \lambda - \beta' \lambda \tag{13}$$

We can perform a Taylor expansion up to second order of $\mathcal{L}(\lambda)$:

$$\mathcal{L}(\lambda) \approx \mathcal{L}(\hat{\lambda}_{\text{MAP}}) + \frac{1}{2} \left. \frac{\partial^2 \mathcal{L}(\lambda)}{\partial \lambda^2} \right|_{\hat{\lambda}_{\text{MAP}}} (\lambda - \hat{\lambda}_{\text{MAP}})^2 \tag{14}$$

We obtain the uncertainty σ^2_{MAP} by evaluating the inverse of the second derivative (Hessian):

$$\sigma^2_{\text{MAP}} = \left[- \left. \frac{\partial^2 \mathcal{L}(\lambda)}{\partial \lambda^2} \right|_{\hat{\lambda}_{\text{MAP}}} \right]^{-1} = \left[\frac{\alpha' - 1}{\hat{\lambda}^2_{\text{MAP}}} \right]^{-1} = \frac{\hat{\lambda}^2_{\text{MAP}}}{\alpha' - 1} \tag{15}$$

Also here we observe that by replacing the definition of $\hat{\lambda}_{\text{MAP}}$ obtained before, we get

$$\sigma^2_{\text{MAP}} = \frac{\alpha' - 1}{\beta'^2}$$

which is the variance of the Gamma distribution as $\alpha' \to +\infty$.

```r
# 1.c Provide an adequate estimate of the uncertainty on lambda
# We check skewness and kurtosis of the posterior distribution to see if a gaussian approximation is reasonable
skewness <- 2 / sqrt(alpha_post)
kurtosis <- 6 / alpha_post
cat(sprintf("The skewness of the posterior distribution is: %.3f\n", skewness))
cat(sprintf("The kurtosis of the posterior distribution is: %.3f\n", kurtosis))

# As skewness and kurtosis are close to 0, we can use a Gaussian approximation to estimate the uncertainty on lambda
var_lambda <- (alpha_post - 1) / (beta_post^2)
cat(sprintf("The variance of the MAP estimate of lambda is: %.3f\n", var_lambda))
cat(sprintf("The MAP estimate of lambda with gaussian approximation is: %.3f +- %.3f\n", map_estimate, sqrt(var_lambda)))
```

```
The skewness of the posterior distribution is: 0.039
The kurtosis of the posterior distribution is: 0.002
The variance of the MAP estimate of lambda is: 0.011
The MAP estimate of lambda with gaussian approximation is: 5.338 ± 0.103
```

2

Using the same above Poissonian model:

2.1

Derive the posterior predictive distribution for a new observation x_{new}.
The definition of the posterior predictive distribution leads to the calculation of the following integral:

$$p(x_{\text{new}}|D) = \int_0^{+\infty} p(x_{\text{new}}|\lambda)p(\lambda|D)d\lambda \tag{16}$$

By substituting the likelihood and the posterior functions we obtain:

$$\begin{aligned} p(x_{\text{new}}|D) &= \int_0^{+\infty} \left(\frac{\lambda^{x_{\text{new}}} e^{-\lambda}}{x_{\text{new}}!}\right) \left(\frac{\beta'^{\alpha'}}{\Gamma(\alpha')}\lambda^{\alpha'-1}e^{-\beta'\lambda}\right) d\lambda \\ &= \frac{\beta'^{\alpha'}}{x_{\text{new}}!\Gamma(\alpha')} \int_0^{+\infty} \lambda^{x_{\text{new}}+\alpha'-1}e^{-(\beta'+1)\lambda}d\lambda \\ &= \frac{\beta'^{\alpha'}}{x_{\text{new}}!\Gamma(\alpha')} \frac{\Gamma(\alpha'+x_{\text{new}})}{(\beta'+1)^{x_{\text{new}}+\alpha'}} \end{aligned} \tag{17}$$

where the last integral is evaluated by recognizing it is the core of an unnormalized Gamma distribution. We can rewrite the result in the following fashion:

$$p(x_{\text{new}}|D) = \frac{\Gamma(\alpha'+x_{\text{new}})}{x_{\text{new}}!\Gamma(\alpha')} \left(\frac{\beta'}{\beta'+1}\right)^{\alpha'} \left(\frac{1}{\beta'+1}\right)^{x_{\text{new}}} \tag{18}$$

This form allows us to recognize that the posterior predictive distribution is a Negative Binomial distribution, which is a probability mass function that predicts the probability of observing k failures before reaching the r-th success in a sequence of IID Bernoulli trials:

$$P(X=k) = \binom{k+r-1}{k} p^r(1-p)^k = \frac{\Gamma(k+r)}{k!\Gamma(r)} p^r(1-p)^k \tag{19}$$

where we exploit the property $\Gamma(n) = (n-1)!$. Therefore, x_{new} corresponds to k failures, while the Negative Binomial parameters correspond to:

$$r = \alpha' \qquad \text{and} \qquad p = \frac{\beta'}{\beta' + 1} \tag{20}$$

```r
# 2.a Posterior predictive distribution for a new observation
# The posterior predictive distribution is a Negative Binomial pmf with parameters r = alpha_post and p = beta_post / (beta_post + 1) where the k failures correspond to x_new

prob_param <- beta_post / (beta_post + 1)
x_max <- qnbinom(0.999, size = alpha_post, prob = prob_param) # sensible maximum
x_vals <- 0:x_max # sequence of integers as we are dealing with a discrete distribution
pmf <- dnbinom(x_vals, size = alpha_post, prob = prob_param)

plot_data <- data.frame(
  x_new = x_vals,
  probability = pmf
)

# plot
ggplot(plot_data, aes(x = x_new, y = probability)) +
  geom_segment(aes(x = x_new, xend = x_new, y = 0, yend = probability), color = "orange", linewidth = 3) +
  labs(title = "Posterior Predictive Distribution",
       x = expression(x[new] ~ "(Number of Events)"),
       y = "Probability Mass") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold"))
ggsave("ex1/posteriorPredictive_Poisson.png", width = 8, height = 6, dpi = 300)
```

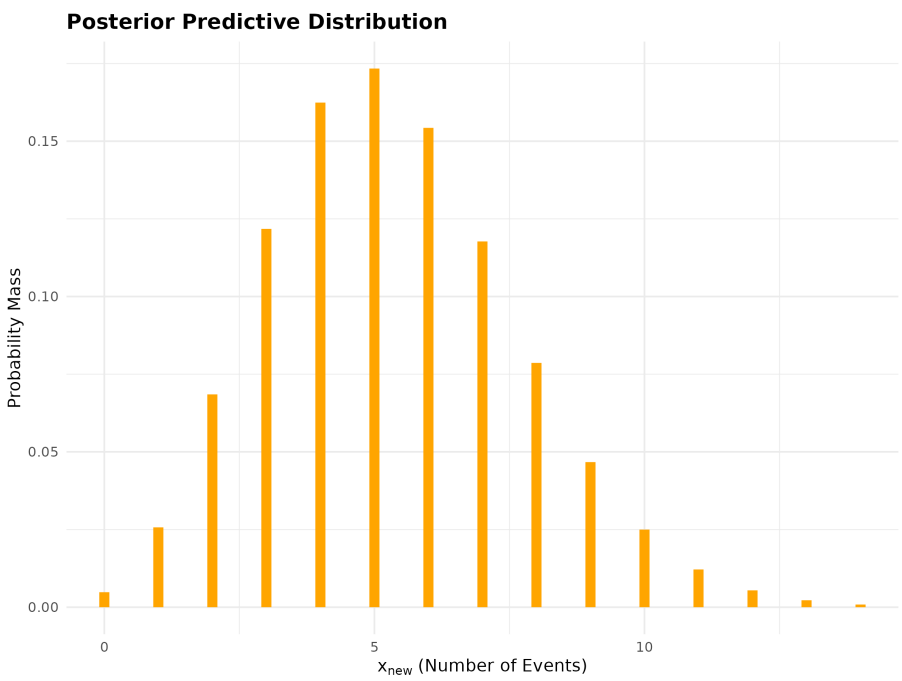

Figure 2: The posterior predictive distribution, which takes the exact form of a Negative Binomial probability mass function. From this distribution, we can obtain the probability of observing a specific number of future events, x_{new}.

2.2

Compute the probability that the next observation exceeds twice the empirical average $\bar{x}$:

$$P(x_{new} > 2\bar{x})$$

To find this probability, we use the Cumulative Distribution Function (CDF) of the Negative Binomial distribution we just derived. The probability of exceeding twice the empirical average is the complement of the probability of observing a value less than or equal to $2\bar{x}$:

$$P(x_{\text{new}} > 2\bar{x}) = 1 - P(x_{\text{new}} \leq 2\bar{x}) = 1 - F_{\text{NB}}(2\bar{x}; r, p) \tag{21}$$

where F_{NB} represents the CDF of the Negative Binomial distribution evaluated at $2\bar{x}$ using our previously defined parameters r and p.

```r
# 2.b Compute the probability that the next observation exceeds twice the empirical average
threshold <- 2 * mean(data$data)
exceeds_prob <- 1 - pnbinom(q = threshold, size = alpha_post, prob = beta_post / (beta_post + 1)
    )
cat(sprintf("The probability that the next observation exceeds twice the empirical average is:
    %.3f\n", exceeds_prob))
```

```
The probability that the next observation exceeds twice the empirical
average is: 0.021
```

3

Consider now an alternative model with a Negative Binomial distribution for the data:

$$x_i \sim NegBin(r, \pi)$$

where $r > 0$ and $0 < \pi < 1$ are the model parameters.

3.1

Give the posterior distribution $p(r, \pi|D)$, motivating your choice of prior distribution.
To derive the joint posterior distribution $p(r, \pi|D)$, we first define the likelihood function for the Negative Binomial model over N independent observations. Dropping the factorial terms that do not depend on the parameters, we have:

$$p(D|r, \pi) = \prod_{i=1}^{N} \frac{\Gamma(x_i + r)}{x_i!\Gamma(r)} \pi^r (1-\pi)^{x_i} \propto \left(\prod_{i=1}^{N} \frac{\Gamma(x_i + r)}{\Gamma(r)} \right) \pi^{Nr} (1-\pi)^{\sum_{i=1}^{N} x_i} \tag{22}$$

where we again denote the sum of observations as $y = \sum_{i=1}^{N} x_i$.
Next, we must choose appropriate priors for the parameters $r \in (0, \infty)$ and $\pi \in (0, 1)$. Assuming prior independence, $p(r, \pi) = p(r)p(\pi)$, we assign the following priors:

- **Prior for π:** Since π represents a probability restricted to the $(0, 1)$ interval, a Beta distribution is the natural choice. We select the Jeffreys prior, $\pi \sim \text{Beta}(\alpha_\pi, \beta_\pi)$, with hyperparameters $\alpha_\pi = 0.5$ and $\beta_\pi = 0.5$. This is a standard, uninformative objective prior that is invariant under parameterization:

$$p(\pi) \propto \pi^{\alpha_\pi - 1} (1-\pi)^{\beta_\pi - 1} \tag{23}$$

- **Prior for** r: Since the size parameter r must be strictly positive, we assign a Gamma prior. We choose weakly informative hyperparameters as done before, $r \sim \text{Gamma}(a, b)$ with $a = 0.01$ and $b = 0.01$, to ensure the distribution is broad:

$$p(r) \propto r^{a-1} e^{-br} \tag{24}$$

Applying Bayes' theorem, the unnormalized joint posterior distribution is proportional to the product of the likelihood and the priors:

$$\begin{aligned} p(r, \pi|D) &\propto p(D|r, \pi)p(r)p(\pi) \\ &\propto \left[\left(\prod_{i=1}^{N} \frac{\Gamma(x_i + r)}{\Gamma(r)} \right) \pi^{Nr} (1-\pi)^y \right] \left(r^{a-1} e^{-br} \right) \left(\pi^{\alpha_\pi - 1} (1-\pi)^{\beta_\pi - 1} \right) \end{aligned} \tag{25}$$

Grouping the terms by parameter, we obtain the final unnormalized joint posterior distribution:

$$p(r, \pi|D) \propto \left(\prod_{i=1}^{N} \frac{\Gamma(x_i + r)}{\Gamma(r)} \right) \pi^{Nr + \alpha_\pi - 1} (1-\pi)^{y + \beta_\pi - 1} r^{a-1} e^{-br} \tag{26}$$

Unlike the Poisson model explored previously, the evidence integral for this Negative Binomial model cannot be evaluated analytically. Furthermore, the joint posterior cannot be factored into a standard and closed-form probability distribution. In the code reported below, we evaluate the unnormalized posterior over a 2D grid to provide a visualization of the parameter space.

```r
# 3.a Consider an alternative model with a Negative Binomial distribution for the data, with parameters size r and prob pi
# Give the posterior distribution for the parameters by choosing appropriate priors

# We assume a Gamma prior for the size parameter and a Beta prior for the prob parameter. The likelihood is given by the Negative Binomial distribution.
# The posterior distribution can be calculated using Bayes' theorem, but it does not have a closed-form solution due to the complexity of the evidence integral.
# However, we can still plot the unnormalized posterior to visualize the distribution of the parameters.
# As probabilities could be infinitesimal, we will evaluate the log of the unnormalized posterior distribution.

# we define negative log-posterior function as later in point 3.b we will use it for optimization
neg_log_post <- function(params, data) {
  r <- params[1]
  pi <- params[2]

  # Log-Likelihood of the Negative Binomial distribution
  log_lik <- sum(dnbinom(data, size = r, prob = pi, log = TRUE))

  # Weakly informative Gamma for r (shape=0.01, rate=0.01)
  log_prior_r <- dgamma(r, shape = 0.01, rate = 0.01, log = TRUE)

  # Jeffreys prior log-Beta for pi (shape1=1/2, shape2=1/2)
  log_prior_pi <- dbeta(pi, shape1 = 1/2, shape2 = 1/2, log = TRUE)

  log_post <- log_lik + log_prior_r + log_prior_pi
  return(-log_post)
}

# We set up the grid for plotting
sample_mean <- mean(data$data)
sample_var <- var(data$data)

```

```r
# This estimated values help us to set a grid that is focused where the posterior is significant
pi_mom <- sample_mean / sample_var
r_mom <- (sample_mean^2) / (sample_var - sample_mean)

r_vals <- seq(max(0.01, r_mom * 0.1), r_mom * 2, length.out = 300) # we use the max as r cannot be zero
pi_vals <- seq(max(0.01, pi_mom - 0.2), min(0.99, pi_mom + 0.2), length.out = 300)
grid <- expand.grid(r = r_vals, pi = pi_vals)

grid$log_post <- apply(grid, 1, function(row) {
  -neg_log_post(c(row["r"], row["pi"]), data = data$data)
})

# Shift maximum to 0 to prevent underflow, then exponentiate to recover the posterior values
grid$posterior <- exp(grid$log_post - max(grid$log_post))

ggplot(grid, aes(x = r, y = pi, z = posterior)) +
  geom_contour_filled(bins = 20) +
  labs(title = "2D Unnormalized Posterior Distribution",
       x = expression("Size Parameter " ~ r),
       y = expression("Probability Parameter " ~ pi),
       fill = "Relative\nDensity") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold"),
        legend.position = "right")
ggsave("ex1/posterior_MAPestimate_NegativeBinomial.png", width = 8, height = 6, dpi = 300)
```

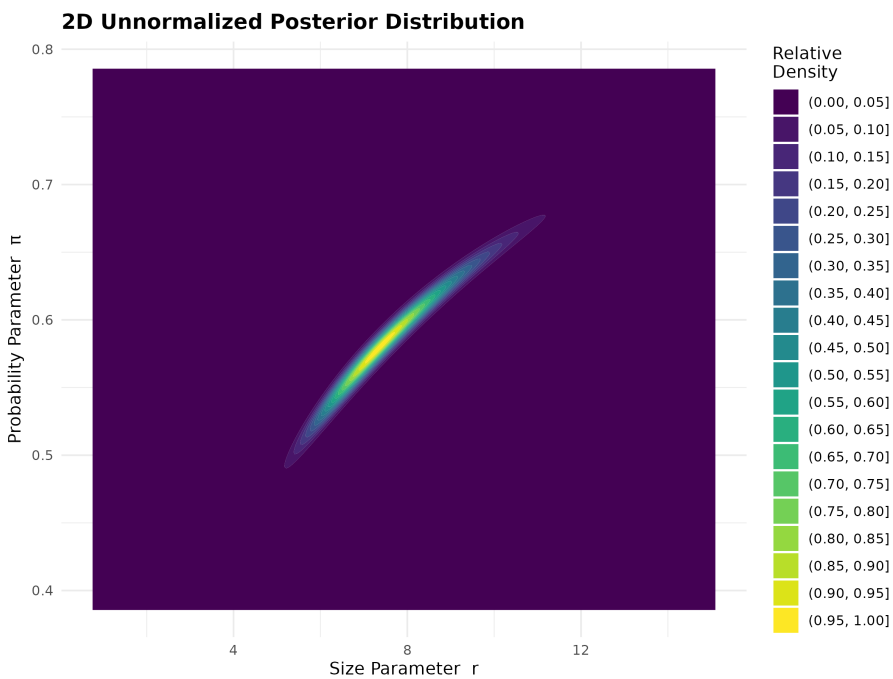

Figure 3: Unnormalized joint posterior distribution for the parameters r and π. The posterior exhibits a distinct, unimodal peak in the parameter space, justifying the use of the Gaussian approximation applied in Section 3.2.

3.2

Assuming a Gaussian shape of the posterior, give a point estimate of the parameters r and π via MAP estimation and its associated error.

Because the joint posterior distribution lacks an analytical solution but presents a unimodal and

peaked shape, we compute the MAP estimates numerically. We achieve this by minimizing the negative unnormalized log-posterior:

$$(\hat{r}_{\text{MAP}}, \hat{\pi}_{\text{MAP}}) = \underset{r,\pi}{\text{argmin}} \left[-\log p(r, \pi | D) \right] \tag{27}$$

To strictly enforce the theoretical boundaries of the parameters ($r > 0$ and $0 < \pi < 1$), we employ a bounded numerical optimization algorithm (L-BFGS-B).
Assuming a Gaussian approximation, we can determine the uncertainties on these point estimates by evaluating the Hessian matrix H at the MAP estimates. The covariance matrix Σ is given by the inverse of this Hessian:

$$\Sigma = H^{-1} = \left[- \begin{pmatrix} \frac{\partial^2 \log p}{\partial r^2} & \frac{\partial^2 \log p}{\partial r \partial \pi} \\ \frac{\partial^2 \log p}{\partial \pi \partial r} & \frac{\partial^2 \log p}{\partial \pi^2} \end{pmatrix} \right]^{-1}_{(\hat{r}_{\text{MAP}}, \hat{\pi}_{\text{MAP}})} \tag{28}$$

The associated standard errors for each parameter are given by the square roots of the diagonal elements of the covariance matrix:

$$\sigma_r = \sqrt{\Sigma_{11}} \qquad \text{and} \qquad \sigma_\pi = \sqrt{\Sigma_{22}} \tag{29}$$

Extracting these values from the computational optimization yields the final point estimates and their associated Gaussian uncertainties:

$$r = \hat{r}_{\text{MAP}} \pm \sigma_r \qquad \text{and} \qquad \pi = \hat{\pi}_{\text{MAP}} \pm \sigma_\pi \tag{30}$$

```r
# 3.2 Assuming gaussian shape, compute MAP estimates and uncertainties for the parameters
# We use numerical optimization to find the MAP estimates and the Hessian to estimate the covariance matrix

# We use "L-BFGS-B" method as we want to enforce boundaries: r > 0 and 0 < pi < 1
initial_params <- c(r_mom, pi_mom)
fit <- optim(par = initial_params,
             fn = neg_log_post,
             data = data$data,
             method = "L-BFGS-B",
             lower = c(1e-5, 1e-5), # we set lower bounds to (r, pi)
             upper = c(Inf, 1 - 1e-5), # we set upper bounds to (r, pi)
             hessian = TRUE)

map_r <- fit$par[1]
map_pi <- fit$par[2]

# find the covariance matrix by inverting the Hessian matrix
cov_matrix <- solve(fit$hessian)

std_r <- sqrt(cov_matrix[1, 1])
std_pi <- sqrt(cov_matrix[2, 2])

cat(sprintf("MAP Estimate for r: %.3f +- %.3f\n", map_r, std_r))
cat(sprintf("MAP Estimate for pi: %.3f +- %.3f\n", map_pi, std_pi))
cat("\nCovariance Matrix:\n")
cat(sprintf("%.3f %.3f\n%.3f %.3f\n", cov_matrix[1, 1], cov_matrix[1, 2], cov_matrix[2, 1], cov_matrix[2, 2]))
```

```
MAP Estimate for r: 7.355 ± 1.140
MAP Estimate for pi: 0.579 ± 0.038
Covariance Matrix:
1.300 0.043
0.043 0.001
```

3.3

Under the same assumption, compute the evidence of the Negative Binomial model.
Because calculating the evidence requires evaluating an integral that lacks an analytical solution, we rely on the Gaussian approximation. By approximating the posterior as a 2D Gaussian around the MAP estimate, the integral of the unnormalized posterior is equal to the peak value multiplied by the volume of the Gaussian:

$$p(D) \approx p^*(\hat{r}_{\text{MAP}}, \hat{\pi}_{\text{MAP}}) \times (2\pi)^{\frac{d}{2}} \sqrt{\det(\Sigma)} \tag{31}$$

where $d = 2$ is the number of parameters and Σ is the covariance matrix.
Because probabilities of large datasets multiply to extremely small numbers, directly calculating the evidence leads to floating point underflow. Therefore, it is necessary to compute the log-evidence instead:

$$\log p(D) \approx \log p^*(\hat{r}_{\text{MAP}}, \hat{\pi}_{\text{MAP}}) + \log(2\pi) + \frac{1}{2}\log(\det \Sigma) \tag{32}$$

In the code implementation, we obtain the final result by substituting our values directly into this logarithmic expression, avoiding exponentiation to prevent numerical underflow.

```r
# 3.3 Evidence in Gaussian approximation

log_evidence_nb <- -fit$value + log(2 * pi) + 0.5 * log(det(cov_matrix))
cat(sprintf("The log-evidence in Gaussian approximation is: %.3f\n", log_evidence_nb))
# cat("The evidence in Gaussian approximation is:", exp(log_evidence), "\n") => underflow!
```

```
The log-evidence in Gaussian approximation is: -1242.164
```

3.4

Compare this result with the evidence obtained for the Poisson model by computing the Bayes factor:

$$\text{BF} = \frac{p(D|M_{\text{NB}})}{p(D|M_{\text{Poisson}})} \tag{33}$$

To evaluate this ratio computationally without encountering numerical underflow or overflow, we must utilize the log-evidences. The logarithm of the Bayes factor is exactly the difference between the two log-evidences:

$$\log \text{BF} = \log p(D|M_{\text{NB}}) - \log p(D|M_{\text{Poisson}}) \tag{34}$$

The final Bayes factor can then be computed by exponentiating this difference:

$$\text{BF} = \exp\left(\log p(D|M_{\text{NB}}) - \log p(D|M_{\text{Poisson}})\right) \tag{35}$$

```r
# 3.4 Calculate the Bayes factor between the two models
log_evidence_Poisson <- alpha_prior * log(beta_prior) +
                        lgamma(alpha_post) -
                        lgamma(alpha_prior) -
                        alpha_post * log(beta_post) -
                        sum(lfactorial(data$data))

BF <- exp(log_evidence_nb - log_evidence_Poisson)
cat(sprintf("The Bayes factor in favor of the Negative Binomial model over the Poisson model is:
     %.3e\n", BF))
```

```
The Bayes factor in favor of the Negative Binomial model over the Poisson
model is: 8.393e+17
```

4

Discussion: Which model is favored and to which degree? Is this unexpected?
With a calculated Bayes Factor on the order of 10^{18}, the Negative Binomial model is overwhelmingly favored over the Poisson model.
At first glance, this might seem to contradict Occam's Razor, which generally favors simpler models with fewer parameters. However, this result is expected due to the assumption that in Poisson model the variance must exactly equal the mean ($E[x] = Var[x] = \lambda$).
Because the sample mean and sample variance in our empirical dataset are different, the Poisson model is fundamentally incapable of providing a good fit. Conversely, the Negative Binomial distribution naturally handles differing means and variances. The introduction of the second parameter provides the flexibility needed to capture the true spread of the data, resulting in a better fit and justifying the Bayes Factor.

```r
# 4. Which model is favored and to which degree? Is this unexpected?
sample_mean <- mean(data$data)
sample_var <- var(data$data)
cat(sprintf("Sample mean: %.3f\n", sample_mean))
cat(sprintf("Sample variance: %.3f\n", sample_var))
```

```
Sample mean: 5.340
Sample variance: 9.119
```